

# Louis RUBLÉ

Seeking a 6-month internship as a Robotics & Perception Software Engineer,  
available worldwide starting August 2025

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 sioulruble |  Louis Ruble |  Personal Website

## EDUCATION

### Master of Engineering in Computer Science

2022-2025

Université de Technologie de Compiègne, France | Specilization in Intelligent Systems and Robotics

**Exchange semester in Zaragoza, Spain** | Master of Robotics, Graphics and Computer Vision

**Relevant courses** : Computer Vision, Autonomous Robots, Machine Learning, Algorithms and Data structures, OOP



### 2-year University diploma in Electrical Engineering and Industrial Computing

2020-2022

IUT de Cachan, France | Specilization in Real-Time Computing and Embedded Networks

**Relevant courses** : Signal Processing, Embedded Systems, Electronics, Control Systems



## EXPERIENCE

### Software Robotics Engineering Intern

Enchanted Tools - Paris | Sep 2023 - Feb 2024

Gained in-depth knowledge of the **robotics stack**, including deployment and workflow management. Developed a **multimodal** platform using facial landmarks, ASR and Speech Source Localization (SSL) in Python to enhance **active speaker detection**. Programmed and integrated new natural **character behaviors** into the robot using Godot, C++.



### Affective Computing Research Assistant

UTC - Compiègne | 2024 - 2025

**SPRING 2024** - Developed an **AR algorithm** simulating VR headsets on faces. Created an **emotion recognition dataset** and implemented a **CNN model** with TensorFlow for VR users emotion recognition.

**SPRING 2025** - Researching state-of-the-art methods for dimensional **emotion analysis**, implementing **Vision Transformer**-based regression model to predict **valence** and **arousal**. Plan to draft a research paper to present findings.



### Image Processing Intern

Laboratoire Signaux & Systèmes - Gif sur Yvette | May 2022 - Aug 2022

Imported an **image reconstruction algorithm** from MATLAB to Python to **improve performance**. Implemented wavelet transforms, gradient functions, and convolution operations in Python. Debugged and optimized the code translation process.



## SKILLS

**Langages** : Python, C/C++, Matlab, Lisp, SQL | **Frameworks / Tools** : Github, ROS, Gazebo, Linux, Qt, PostgreSQL

**Libraries** : Numpy, Sklearn, OpenCV, Tensorflow/Pytorch, Matplotlib, Pandas

## INTERESTS

**Cinema / Scifi**: Active member of the UTC cinema association

**Chess**: Competitive practice (~2150 Elo rating, top 15 FR U18)

**Combat sports**: Amateur Boxing and Sanda

## LANGUAGES

**French** - Native

**English** - Technical language practice

**German / Spanish** - Beginner